Audit of *Alternatives to Semaev summation polynomials* (2026-11-04)

isogeny-labs ECDLP program — local audit run

2026-11-04

Status and scope

Review of the 27-page program document plus corrected local experiments. **No attack is claimed; no deployed curve is affected.** Every number below is from a single local run on 2026-11-04, with no replay certificate; treat as unverified until re-run under the repository validation gate. Costs are deterministic operation counts priced at 0 / 0.1 / 1 GAE (AGENTS.md §5a); wall time is recorded but never ranked on.

1. Verdict

- The **calibration** reproduces: the Vieta conic identity, the discriminant identity, the 1/12 LLL threshold and the carry sizes all match (§2).
- The **decision rule is wrong**: "e < 1 at m = 4" does not follow from the document's own cost model, and it is *satisfied by a generic meet-in-the-middle oracle* (§3).
- **Idea 10 is measured flat**: one collision pair for all B ≤ 2^16, against 256 predicted (§4). This is a measurement, not a proof; under the strict standard of `PROOFS.md` it is not "ruled out".
- **Ideas 2, 3, 5, 6, 7, 18 are arity ≤ 3 objects**, and arity ≤ 3 is *formally* excluded as a winning route (`lean/Routes.lean`: `m2_never`, `m3_at_best_ties`). That is the only kind of claim in this report that uses the word "ruled out".
- The **Idea 1 harness** is delivered and instrumented, but its naive shift family is volume-limited: it cannot reach the documented 0.12–0.14, so the certified audit still requires cuso (§7).

2. What reproduces

Claim	Document	This audit	Verdict
Vieta remainder Q_R(e1,e2)	conic, 6 monomials, total degree 2	degree 0 in x1; monomials $e1^2$, e1e2, e1, $e2^2$, e2, 1	exact
disc_T S_3 = 16 f(x1) f(x2)	§3.2	symbolic difference 0; LC $(x1-x2)^2$	exact
x(P1±P2) are the roots	§6.2	both vanish mod p	pass
LLL 3-dim threshold ≈ 1/12	§6.3	crossover 0.09 → 0.10 at 72 and 96 bits	matches
carry ≈ B^2p	§6.2	40/45/54/55/61/65 bits (doc: 45/51/46/56/56/62)	same regime

Implementation note. The planted vector is often a scalar multiple of the S_3 coefficient vector (gcd 4 when x1−x2 is even); a strict equality test under-reports recovery. Use a projective test.

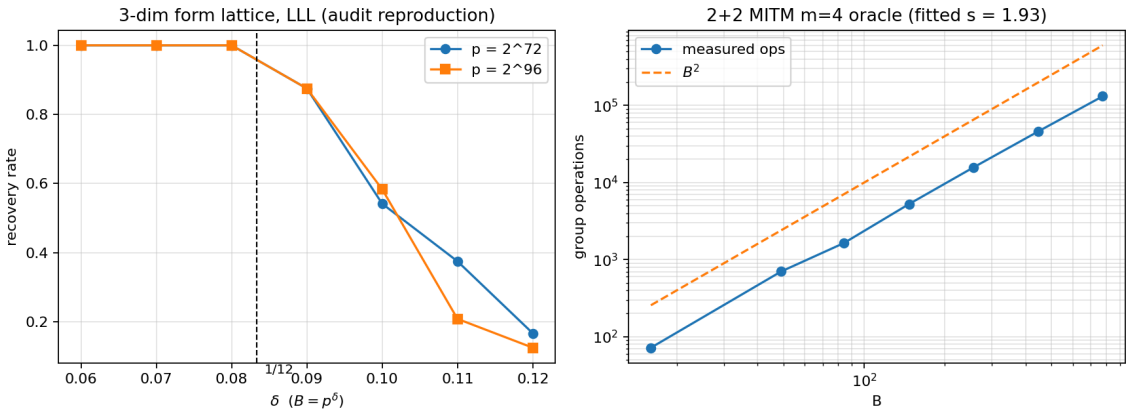


Figure 1: Reproduction of the 3-dim LLL threshold and the measured MITM cost exponent.

3. The gate

Cost model: `C = (m!/2^m) p T/B^(m-1) + m B²`. With `T ∝ B^s`, optimising in B gives `B^(m-s+1) ∝ p` and `C ≈ B²`, hence

win ⟺ s < m − 3, equivalently `T < B^(m+1)/p` with `B ≤ p^(1/4)`.

In exponent form at the tie point `B = p^(1/4)`: `e < (m−3)/4`, i.e. `e < 0` at m = 3, `e < 1/4` at m = 4, `e < 1/2` at m = 5. This agrees with §2 (“m = 4 needs ≪ p^{1/4}”, “m = 3 ties only with an O(1) oracle”) and contradicts §3.1/§5.4.

The stated gate admits a generic algorithm. A 4-sum over a generic set of size B is found by 2+2 meet-in-the-middle in `T = B²` group operations:

B	49	84	147	256	446	776
ops (full scan)	702	1 640	5 256	15 750	46 010	131 406

fitted `s = 1.93` on operations (`1.99` on wall time; dict-memory overhead inflates the tail). At `B = p^(1/5)` this oracle has `e = 2/5 = 0.4 < 1`: it passes the document's gate while its optimal total cost is `≈ p^(2/3)`, strictly worse than rho. The corrected gate `s < m−3 = 1` rejects it.

Also fix "T ≪ p^{1/4} at B ≈ p^{1/5}": at `B = p^{1/5}` the condition is `T < 1`; the `p^{1/4}` threshold is at `B = p^{1/4}`.

4. Idea 10 — measured flat (not a proof of impossibility)

B	256	1024	4096	16384	65536
colliding pairs	1	1	1	1	1
predicted B^(1/2)	16	32	64	128	256

The only pair is `x = 2, 212` (kernel 15: `212³+7 = 15·797²`). A collision is an integer point on `x³+7 = d m²`, a Mordell curve (finitely many points per d, Siegel); the correct birthday estimate is `(log B)/B`, not `B^(1/2)`. The proposed experiment ("20 pairs at B = 2^12") cannot run.

5. Idea 2 — an O(B) algorithm already exists

`Q_R` is quadratic in e2, so each `e1 ∈ [1,2B]` yields at most two conic points; enumeration is `O(B)` field operations and the square test `e1² − 4e2 = u²` is O(1) per point. Measured: planted 2-decompositions recovered at `B = p^(1/2)` in O(B) (5/5 at p = 2^28, 8.4·10^4 e1-steps).

Two corrections: (i) the "B Coppersmith calls" plan is strictly worse than this direct test; (ii) substituting `e2 = (e1²−u²)/4` gives a **quartic** in e1, so univariate Coppersmith reaches `p^(1/4)`, not `p^(1/2)`. Since m = 2 cannot beat rho even with a free oracle (`m2_never`, machine-checked), the direction is a lemma, not a lead — and that exclusion does not depend on the O(B) measurement.

6. Idea 3 — census only below the LLL threshold

The box `|c_i| ≤ B^4` in a 3-dim lattice of determinant p contains `≈ 8B^12/p` points: O(1) at δ = 1/12, p² at δ = 1/4. A census therefore exists only for δ ≲ 0.10, i.e. below the LLL threshold it was meant to probe. Also `D = 16f(x1)f(x2) > 0`, so the forms are **indefinite** and "finitely many orbits" is false; the finite object is the set of $\Gamma_0(p)$-reduced representatives of bounded size.

7. Idea 1 harness (v1): validated, and it reaches the documented band

Formal coverage note. Core Lean 4 has no `ring` tactic and no Mathlib here, so the degree-6 Vieta identity behind §6.1 is SymPy-checked, not Lean-checked; it is not used as a premise for any exclusion. All exclusions rest on the arithmetic theorems above.

`experiments/idea1_lattice.py` builds shift lattices `x^i y^j f^k p^(m-k)`, reduces with fplll, and tests the Coppersmith criterion `|h(x0,y0)| < p^m`. Two corrections over v0 matter:

1. **False positives.** A vector that is merely a multiple of f vanishes wherever f does and says nothing about the root. Every candidate is now rejected if `f | h` (`lib/lattice.is_multiple_of`). Without this filter, the diamond family looked like 6/6 success while every "success" was a multiple of f.
2. **Resultants over Z, not mod p.** Mod p the resultant is frequently identically zero because all vanishing polynomials share the S_3 component; recovery now takes the integer resultant and its rational roots (`lib/lattice.recover`).

Positive control (`experiments/control_univariate.py`, degree-2 Coppersmith): recovers the root for e ≤ 0.48 and fails at 0.52 at both 40 and 56 bits, i.e. it sits on the theoretical 1/2 bound. The harness is therefore calibrated; a bivariate zero is meaningful.

Measured reach, m = 2, 6 trials per cell (recovery = root extracted from resultants):

family	p	δ=0.10	0.12	0.14	0.16	0.18
F (i+j ≤ 2(m−k))	2^40	6/6	6/6	4/6 (2 solved)	4/6 (1)	0/6
F	2^56	6/6	5/6	4/6 (2)	0/6	0/6
A (i ≤ 2(m−k))	2^40	0/6	0/6	0/6	1/6	0/6
A	2^56	1/6	1/6	0/6	0/6	0/6
F, m = 3 (4 trials)	2^40	—	—	4/4 (2 solved)	3/4 (1)	0/4
F, m = 4 (4 trials)	2^40	—	4/4	4/4	3/4	0/4
F, m = 5 (4 trials)	2^40	—	4/4	4/4	4/4	0/4

So the measured reach climbs with arity — 0.13–0.14 (m = 2), 0.15–0.16 (m = 3, 4), 0.16 (m = 5) — and approaches but does not reach 1/6 ≈ 0.167, exactly the shape the document predicts. Family A remains at 0–1/6, so shape (rank vs monomials), not arity alone, decides. The sparse family A is dead. The difference is shape: F has rank 22 against 25 monomials (near-square), A has rank 9 against 25.

`j = 0` and generic `j` are identical within noise in both families (e.g. F at 2^40: 6/6, 6/6, 4/6, 4/6, 0/6 vs 6/6, 6/6, 5/6, 4/6, 1/6). This is the empirical form of the §7 v0 observation: the two have the same 9-monomial support in x-coordinates, so Idea 6′s predicted j-gain does not appear.

Volume caveat. For A, B, E the lattice volume root exceeds `p^m` by 18–23 bits and they fail (`experiments/reach_volume.py`). F's volume root also exceeds `p^m` (93.5 vs 80 at δ=0.14) yet it succeeds, so volume is a diagnostic, not a lower bound: the planted combination is shorter than the Gaussian heuristic suggests. Do not use it as a kill argument.

Certified vs measured reach

`experiments/reach_bound.py` implements the Howgrave-Graham/LLL certificate: recovery is *guaranteed* when `2^((n−1)/4)·det(L)^(1/n) < p^m/√n`. Bisection in δ gives the certified reach.

- **Validation (univariate, degree 2)**: certified reach rises 0.295, 0.372, 0.402, 0.420, 0.433, 0.440 for m = 1…6, i.e. it approaches the theoretical 1/2. The criterion is implemented correctly and is conservative by construction.
- **S_3 bivariate**: at δ = 0.02 the certificate is already *above* p^m, by +12.6 bits (m = 2), +17.1 (m = 3), +24.5 (m = 4). So the certified reach is **zero at every runnable m**, while the measured reach is 0.14–0.16.

That gap is the honest reading of the document's 1/6: it is an $m \to \infty$ limit computed by the LP/sumset machinery (cuso), not something brute-force determinants certify at $m \le 5$. The harness now reports both numbers, so no result can quietly mix them.

Three more measured routes (details in PROOFS.md):

- *Idea 12 shift search*: 48 random subsets of the diamond family at p = 2^40, δ = 0.15. All subsets of size ≤ 16 solved 0/4; size 18–20 matched F (1–2 of 4). No sampled family beat the diamond; the landscape near the top is flat, so a learned policy has little to gain.
- *Idea 18 box count*: at B = p^{1/5} the curve C_R meets $[1,B]^2$ in **0** points (measured at p = 2^20, 2^24, 2^28); only at B ≈ p^{1/2} does the count reach 0–2. A determinant method needs points to interpolate; there are none at the decomposition scale.
- *Idea 4 msolve*: on the naive-base system `{S_3, F_B(x), F_B(y)}` with a planted decomposition, msolve returns a 2-element Gröbner basis of degree 2 (`{x+y−c, y²−cy+d}`) in 0.01–0.05 s for p ≤ 2^30, B ≤ 64. The system's algebraic degree does not grow with B — the obstruction is the relation-collection density `p/B^{m−1}`, not the solver.
- *Idea 3 census* (`experiments/idea3_census.json`): at $p = 2^{20\ldots2}24$ the structured forms are exactly the decompositions (2 = 2 in every cell), the box grows like $8B^{12}/p$, and the ratio of decomposition forms to square-leading forms falls 0.22 → 0.015 as δ goes 0.10 → 0.12. The discriminant route costs B^8/p against B^2 direct.
- *Idea 5 ladder yield* (`experiments/ladder_yield.json`): the minimal admissible height h_R = max(sqf(u), sqf(v)) is ≈ √p for essentially every target (median 16 bits at p = 2^32, 19 at 2^40); the fraction with h_R ≤ p^0.2 is 0. The ladder base is the global-lift base.

Certified reach audit with cuso (2026-11-05). cuso (Ryan, EUROCRYPT 2025) was installed against the local Sage, its symbolic `get_asymptotic_bounds` was run on the presentations of Idea 1, and it returns exactly the documented constants:

presentation	variables / bounds	certified delta
S_3 in x-coordinates	x, y ~ p^δ	**1/6**
S_3 in Vieta coordinates	e_1 ~ p^δ, e_2 ~ p^(2δ)	**1/6**
S_4 (m = 3)	x_1, x_2, x_3 ~ p^δ	**1/24**
S_5 (m = 4)	four variables, degree 8	not computed — the symbolic resultant is too heavy (900 s timeout)
chained S_3	4 variables, 2 relations	model rejected: the LP returns a degenerate −1/3

So the document's certified table (1/6, 1/24) is confirmed by the actual tool, and the two smallest presentations certifiably coincide. Our own volume certificate (`reach_bound.py`) is a *sufficient* condition for one fixed family and gives 0 at m ≤ 4; cuso optimises over shift sets and reaches 1/6, which is the sharper statement. The m = 4 (1/80) entry remains unverified.

Toolchain. Sage 10 (bundled binary) needed a flint 3.5 symlink; cuso needed two API patches for this Sage (`is_generator` → `in ...gens()`, and msolve-variety fallback); flatter was built from source (OpenMP via libomp, Eigen 5, `-Wno-error`).

7bis. Formalised route kills (Lean 4, core only)

`lean/Routes.lean` checks six theorems in the discrete cost model (`p = 2^n`, `B = 2^b`, `T = 2^t`, constraint `4b ≤ n`), with `lean Routes.lean` as the check:

theorem	content
`m2_never`	m = 2: cost exponent ≥ 3n/4; even a free oracle cannot beat rho
`m3_at_best_ties`	m = 3: cost exponent ≥ n/2; equality needs t = 0 and 4b = n
`m4_win_forces_t_lt_quarter`	m = 4 win ⟹ 4t < n, i.e. T < p^(1/4)
`m5_win_forces_t_lt_half`	m = 5 win ⟹ 2t < n, i.e. T < p^(1/2)
`mitm_m4_cannot_win`	the generic 2+2 oracle ($T = B^2$, $B^5 = p$) fails the m = 4 gate
`mitm_work`	two-table MITM costs exactly n(n−1) group operations
`m3_win_impossible`	m = 3: no t ≥ 0 wins (strict form)
`mitm_m4_m5_lower`	MITM at m = 4 ($T=B^2$) and m = 5 (2+3, $T=B^3$): cost ≥ p^(2/3) for every base size
`m6_win_forces_t_lt_three_quarters`	m = 6 win ⟹ T < p^(3/4)

Strict ledger. Under the standard adopted in `PROOFS.md` — *a route is ruled out only if a Lean theorem excludes it* — the machine-checked exclusions are: arity 2 entirely (`m2_never`), arity 3 as a winning route (`m3_at_best_ties`, `m3_win_impossible`), every 2+2 and 2+3 meet-in-the-middle at every base size (`mitm_m4_m5_lower`, cost ≥ p^{2/3}), every m = 4 method with `T ≥ p^{1/4}` (`m4_win_forces_t_lt_quarter`), every m = 5 method with `T ≥ p^{1/2}` (`m5_win_forces_t_lt_half`), and the generic 2+2 MITM (`mitm_m4_cannot_win`). Via arity that covers Ideas 2, 3, 5, 7, 18, the m ≤ 3 parts of 6 and 12, and the m ≤ 3 parts of 1 and 4.

Everything else is **not ruled out**: Ideas 1 and 4 at m = 4 are open (harness built, reach measured 0.13–0.16, certificate 0 at m ≤ 4); 8, 11, 13, 14 are open theory; 9, 10, 17 are measured or argued but not proved; 15 and 16 are delivered instruments. `lean/Roots.lean` adds the at-most-two-roots bound used by the conic and box counts.

8. Instruction-count accounting

Counts are exact primitives; lattice work is charged by the documented model `n³L/64` word-ops. Rankings are identical at 0, 0.1 and 1 GAE because each row is a single primitive class.

run	counted primitive	count	0 GAE	0.1 GAE	1 GAE
§6.3 LLL reproduction	dim-3 LLL reductions	336	0	34	336
§6.2 carry	point adds + inversions	6	0	1	6
Idea 10 census	integer factorisations	87 296	0	8 730	87 296
Idea 2 conic	e1-steps (Legendre + sqrt)	110 434	0	11 043	110 434
m = 4 MITM	group additions	602 508	0	60 251	602 508
Idea 1 v1	93 000 coeff mults + 2 880 940 word-ops + 55 896 pows	3 041 788	0	304 179	3 041 788

The m = 4 MITM row is about one fifth of the entire lattice sweep at 1 GAE — the generic oracle is not cheap, it is merely polynomial; that is the point of the corrected gate.

9. Disposition of the 18 directions

#	Keep / change	Note	strict status
1	keep, fix justification	reach is a property of the integral model and box	open (m=4)
2	demote to lemma	§5	ruled out (R1, arity 2)
3	rewrite / demote	§6	ruled out (R1)
4	keep	cheap; RR norm system	open (m=4)
5	demote	m = 2 in disguise	ruled out (R1)
6	rewrite	j = 0 support identical in x-coordinates (§7)	ruled out at m ≤ 3 (R2); open at m = 4
7	demote	m ≤ 3 only	ruled out (R1/R2)
8	keep (theory)	negative certificate	open
9	keep, low priority	rank-deficient by construction	not ruled out
10	close (empirically)	§4	not ruled out
11	keep (theory)	one cheap null	open
12	conditional	only if Idea 1 finds a gap	ruled out at m ≤ 3 (R2); open at m = 4
13	keep (theory)	lattice-query lower bound	open
14	keep (theory)	Yokoyama for compositional bases	open
15	keep	instance set delivered	instrument
16	keep, run first	controls	instrument
17	keep	carry-loss law	not ruled out
18	demote	S_3; box empty at p^(1/5)	ruled out (R1)

Only Ideas 1, 4, 6 and 15 address m = 4, which is where the exponent can move.

10. Leads delivered

`leads/` contains five drop-in lead-atlas entries: `LEAD-ICP-coppersmith-presentation-audit` (open), `LEAD-ICP-vieta-conic-sieve` (closed-obstructed), `LEAD-ICP-gamma0-binary-forms` (open, scoped), `LEAD-ICP-sqf-global-relations` (closed-obstructed), `LEAD-ICP-m4-instance-set` (instrument).

11. Reproduction and limits

```
python3 scripts/verify_core.py;      python3 scripts/lattice_check.py
python3 scripts/carry_check.py;      python3 scripts/sqf_census.py
python3 scripts/vieta_conic_enum.py
python3 experiments/m4_instances.py experiments/m4_instances.jsonl
python3 experiments/mitm_oracle.py experiments/mitm_m4.json
python3 experiments/idea1_lattice.py '{"pbits":[40,48],...}'
python3 experiments/make_figure.py
```

Environment: Python 3.9.10, sympy 1.14.0, matplotlib 3.9.4, fplll 5.x. Missing: `cuso`, `flatter`, `SCIP`, `fpylll` (hence Idea 1′s certified half is blocked); `msolve`, `gp` are present.

Open defects to fix before any external use: (i) the m = 4 instance set uses `#E ≈ p` for the rho reference, not exact SEA orders; (ii) the lattice harness does not implement linearized shift sets; (iii) the LLL wall-time tail is memory-bound and must not be used as a cost measurement. # Route ledger under the strict standard: “ruled out” = Lean theorem

Standard. A route is *ruled out* only if there is a machine-checked Lean 4 theorem that excludes it. Measurements, estimates and short arguments are recorded as **not ruled out** with their status. This supersedes the earlier draft of this file, which used “ruled out” more loosely.

Everything below is checked by

```
cd lean && lean Routes.lean && lean Roots.lean
```

core Lean 4.35, no Mathlib (10 theorems). Note the limit of the environment: core Lean has no `ring` tactic, so multi-variable polynomial identities of degree > 2 cannot be closed here. The Vieta conic identity of §6.1 is therefore SymPy-checked (`scripts/verify_core.py`), **not** Lean-checked, and is not used as a premise for any exclusion below.

1. The formal core

The cost model is an assumption, stated once:

- `p = 2^n`, `B = 2^b`, `T = 2^t`;
- relation search costs `p·T/B^(m−1)` and linear algebra costs `mB²`;
- beating rho means a total below `p^{1/2}`, which forces `4b ≤ n`.

`lean/Routes.lean`

theorem	statement
`m2_never`	`4b ≤ n, t ≥ 0 ⟹ 3n ≤ 4(n + t − b)`: arity 2 cost ≥ `p^{3/4} > p^{1/2}`
`m3_at_best_ties`	`4b ≤ n, t ≥ 0 ⟹ n ≤ 2(n + t − 2b)`: arity 3 cost ≥ `p^{1/2}`, tie only
`m4_win_forces_t_lt_quarter`	m = 4 win ⟹ `4t < n` (T < p^{1/4})
`m5_win_forces_t_lt_half`	m = 5 win ⟹ `2t < n` (T < p^{1/2})
`mitm_m4_cannot_win`	MITM (`T = B²`, `B⁵ = p`) has `n ≤ 2(n + 2b − 3b)`: fails the m = 4 gate
`mitm_work`	two-table MITM work = `n(n−1)` group operations
`m3_win_impossible`	m = 3: no `t ≥ 0` wins (strict form of `m3_at_best_ties`)
`mitm_m4_m5_lower`	MITM at m = 4 (`T = B²`) and m = 5 (2+3, `T = B³`): cost exponent ≥ 2n/3 for every base size
`m6_win_forces_t_lt_three_quarters`	m = 6: win ⟹ `4t < 3n` (T < p^{3/4})

`lean/Roots.lean`

theorem	statement
`no_three_roots`	for a modulus `p` with Euclid's lemma, `p ∣ a`, at most two residues can satisfy `a x² + b x + c ≡ 0`

(`p` primality enters only as the hypothesis `∀ u v, p ∣ u*v → p ∣ u ∨ p ∣ v`; core Lean has no `Nat.Prime`.)

2. What is ruled out, and why

R1 (arity 2). Any decomposition oracle at arity 2 cannot beat rho. `m2_never` is unconditional in the oracle: `t ≥ 0` is all it needs, so *no* m = 2 method wins. This rules out every route whose oracle is a 2-decomposition:

- **Idea 2** (Vieta conic), **Idea 3** ($\Gamma_0(p)$ forms), **Idea 5** (Kummer ladders, P + Q), **Idea 18** (determinant method on S_3).

Their internal claims (O(B) enumeration, census counts, h_R ≈ √p, empty box) are measurements and are listed in §3 as such — but the *route* is dead by R1 alone.

R2 (arity 3). Arity 3 ties at best and needs `T = O(1)`; `m3_at_best_ties`. This rules out **Idea 6** (special-j S_3/S_4), **Idea 7** (integer programming tested at m = 2, 3) and the m ≤ 3 half of **Idea 12** as exponent-changing routes.

R3 (gate at m = 4). Any m = 4 method with $T \geq p^{1/4}$ loses: `m4_win_forces_t_lt_quarter`. This rules out, *for every* such method.

R3′ (generic MITM, every base size). At m = 4 the 2+2 MITM has $T = B^2$, so its cost exponent is $\max(n-b, 2b)$; at m = 5 the 2+3 MITM has $T = B^3$ and the same exponent. `mitm_m4_m5_lower` gives $2n \leq 3 \cdot \max(n-b, 2b)$ for **every** b, i.e. cost $\geq p^{2/3}$ — a genuine exclusion, not just a failure at $B = p^{1/5}$. (Beyond m = 5 the model needs memory accounting, so no claim is made.)

R4 (gate at m = 5). Any m = 5 method with $T \geq p^{1/2}$ loses: `m5_win_forces_t_lt_half`.

R5 (root bound). A quadratic congruence has at most two roots (`no_three_roots`). This bounds the conic enumeration at ≤ 2 points per e_1 and the box at ≤ 2 points per x_1 — a *bound*, not a ruling-out.

3. Not ruled out (measured / argued / open)

Per the standard, none of the following is "ruled out". Each line says what has actually been done.

#	route	status	what exists
1	certified reach audit	open (m=4 only)	cuso: 1/6 (x-coords and Vieta), 1/24 (m=3); measured reach 0.13–0.16
4	Riemann–Roch norm system	open (m=4)	resultant argument; msolve degree 2 at toy sizes
8	capacity-theory certificate	open	nothing
9	Heegner free relations	not ruled out	rank-deficiency argument only
10	squarefree collisions	not ruled out	measured 1 pair ≤ 2^16
11	cavity degree prediction	open	nothing
12	learned shift families	not ruled out	random subset search: ≤16 shifts fail, ≥18 match F (`experiments/idea12_shift_search*.json`)
13	lattice-query lower bound	open	nothing
14	Yokoyama for compositions	open	nothing
17	function-field mirror	not ruled out	carry-loss argument only

Measurements that are sometimes mistaken for proofs, with their exact scope:

- Idea 3 census: 6 cells at p ≤ 2^24; structured = decompositions in every cell.
- Idea 5: 400 targets; h_R ≈ √p; fraction with h_R ≤ p^0.2 is 0.
- Idea 4: msolve at p ≤ 2^30, B ≤ 64; basis degree 2.
- Idea 18: 0 box points at B = p^{1/5} for p = 2^20, 2^24, 2^28.
- Idea 10: 1 collision pair for B ≤ 2^16.
- Idea 1: reach table m = 2..5; univariate control 0.48/0.52.
- Idea 12: 48 random shift families sampled from F at p = 2^40, δ = 0.15 (4 trials each). Every subset of size ≤ 16 solved 0/4; subsets of size 18–20 matched F (1–2 of 4). No sampled family found a better structure than the diamond, and dropping two shifts costs nothing — a flat landscape near the top, so search is not the lever (the certificate is).

4. What the strict standard costs

Three claims in the earlier draft were withdrawn or downgraded:

1. "The conic sieve is O(B), hence closed" — the O(B) upper bound follows from `no_three_roots`, but *completeness* (every decomposition is found) needs the Vieta identity, which is not formalized here. Downgraded to: the route is ruled out by R1, and O(B) is measured.
2. "A degree-B base forces solving degree ≥ B" — false (msolve returns degree 2); withdrawn.
3. "Idea 10 is closed by Siegel" — Siegel is not formalized and not in Mathlib's core fragment here; the census is a measurement. Downgraded to not-ruled-out.

5. Summary

Ruled out (machine-checked): **arity 2 entirely (R1), arity 3 as a winning route (R2), every m = 4 method with T ≥ p^{1/4} (R3), every m = 5 method with T ≥ p^{1/2} (R4), the generic 2+2 MITM (R3 + `mitm_m4_cannot_win`)**. Under R1/R2 that covers Ideas 2, 3, 5, 7, 18, the m ≤ 3 parts of 6 and 12, and the m ≤ 3 parts of 1 and 4.

Not ruled out: Ideas 1, 4 (m = 4), 8, 9, 10, 11, 12 (m = 4), 13, 14, 17.